

Retail Sales Data Analysis Report

1. Introduction

Retail businesses generate large volumes of sales data that can be analyzed to understand customer behavior, product performance, and overall business trends. This project focuses on performing Exploratory Data Analysis (EDA) on a retail sales dataset to extract meaningful insights and support decision-making.

2. Objective

- To load and clean the retail sales dataset
- To perform descriptive statistical analysis
- To analyze sales trends over time
- To study customer demographics and product performance
- To visualize key insights using charts and graphs
- To provide actionable business recommendations

3. Tools and Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn

4. Dataset Description

Attribute Name	Description
Date	Transaction date
Customer_ID	Unique customer identifier
Product	Product name
Category	Product category
Sales	Sales amount
Quantity	Number of items purchased
Gender	Customer gender (if available)

5. Data Cleaning

- Missing values were removed or handled
- Duplicate records were eliminated
- Data types were corrected (especially date format)

- Invalid or inconsistent entries were filtered

6. Descriptive Statistics

Basic statistical measures such as mean, median, mode, and standard deviation were calculated to understand the distribution and variability of the dataset.

7. Time Series Analysis

Sales data was analyzed over time by grouping transactions by date/month and visualizing trends using line plots.

Observations: Sales showed fluctuations over time, certain periods had higher sales, and peak sales periods were identified.

8. Customer Analysis

Customer purchasing behavior was analyzed by calculating total spending per customer and identifying high-value customers.

Observations: A small group of customers contributed significantly to total sales, and spending patterns varied.

9. Product Analysis

Product performance was evaluated by analyzing sales by product and category.

Observations: Few products generated most revenue while some categories underperformed.

10. Data Visualization

- Line Plot – Sales trend over time
- Bar Chart – Sales by category/product
- Heatmap – Correlation between variables

11. Key Insights

- Sales show seasonal patterns
- Limited products drive most revenue
- High-value customers contribute significantly
- Some categories need improvement

12. Recommendations

- Focus marketing on top-performing products
- Provide personalized offers to high-value customers
- Improve or promote low-performing products
- Increase inventory during peak periods
- Use customer segmentation for targeted marketing

13. Conclusion

This project successfully analyzed retail sales data using EDA techniques. The insights can help improve decision-making, optimize strategies, and enhance customer satisfaction.

